

Module 8 — Geant4 Install & B1 Example

Supplementary to the sPHENIX Standalone Course

1. What Geant4 does (and doesn't)

Geant4 propagates particles through matter. Given a geometry, a physics list, and a primary particle, it simulates step-by-step transport with multiple scattering, energy loss, secondary production, decays, and hadronic interactions. It does **not** generate collisions itself — that's Pythia's job. And it doesn't do reconstruction — that's your job.

2. Download and build

```
cd $HOME/src
wget https://gitlab.cern.ch/geant4/geant4/-/archive/v11.2.1/geant4-v11.2.1.tar.gz
tar xzf geant4-v11.2.1.tar.gz
mkdir geant4-build && cd geant4-build
cmake ../geant4-v11.2.1 \
  -DCMAKE_INSTALL_PREFIX=$HOME/opt/geant4 \
  -DGEANT4_USE_QT=ON \
  -DGEANT4_USE_OPENGL_X11=ON \
  -DGEANT4_INSTALL_DATA=ON \
  -DGEANT4_INSTALL_EXAMPLES=ON \
  -DGEANT4_BUILD_MULTITHREADED=ON
make -j$(nproc)
make install
```

3. The environment

Always source the environment before using Geant4:

```
source $HOME/opt/geant4/bin/geant4.sh
echo $G4LEVELGAMMADATA # sanity check — should point to data dir
```

Note: geant4.sh sets ~15 environment variables pointing to data files (photon evaporation, neutron HP, etc.). Geant4 will initialise without them but physics will be wrong — always source before running.

4. Data files

Variable	Used for
G4LEVELGAMMADATA	Photon evaporation / nuclear levels
G4RADIOACTIVEDATA	Radioactive decay
G4NEUTRONHPDATA	High-precision neutron transport
G4LEDATA	Low-energy EM
G4PARTICLEXSDATA	Hadronic cross sections
G4ENSDFSTATEDATA	ENSDF nuclear levels
G4REALSURFACEDATA	Optical surface properties
G4INCLDATA	INCL++ intranuclear cascade

5. The B1 example — structure

File	Role
B1DetectorConstruction	Geometry: world + target
B1ActionInitialization	Register run/event/tracking actions
B1PrimaryGeneratorAction	Particle gun: proton 6 MeV
B1RunAction	Dose accumulation across run
B1EventAction	Per-event dose bookkeeping
B1SteppingAction	Collects energy deposit per step
exampleB1.cc	main()

6. Running the example

```
cp -r $G4INSTALL/examples/basic/B1 ./B1
mkdir B1-build && cd B1-build
```

```
cmake ../B1  
make  
./exampleB1 run1.mac
```

Open the GUI version (./exampleB1 with no args), then type /run/beamOn 10 in the terminal at the bottom. A visualization window opens showing the tracks.

7. What B1 teaches you

Even though the geometry is trivial (a gamma source and an envelope), B1 exercises every class you'll need in your own simulation: `DetectorConstruction`, `PrimaryGeneratorAction`, `PhysicsList`, `UserRunAction`, `UserEventAction`, `UserSteppingAction`. Read its code line by line before writing your own.

8. Macro anatomy

```
# init.mac
/run/numberOfThreads 4
/run/initialize
/vis/open OGL
/vis/drawVolume
/vis/scene/add/trajectories smooth
/vis/scene/endOfEventAction accumulate 100
/gun/particle proton
/gun/energy 6 MeV
/run/beamOn 100
```

You can run entirely from a macro in batch mode, or start in GUI mode and type commands live.

9. First customisation

Before touching geometry, change the primary particle and watch how dose changes:

```
/gun/particle e-
/gun/energy 20 MeV
/run/beamOn 1000
```

Note: Dose per primary is a few nGy — scale to an expected irradiation. Compare with your analytic estimate before declaring the simulation 'working'.

10. Common build & runtime failures

Symptom	Cause / fix
Qt linker errors on CMake	<code>qtwebengine5-dev</code> missing or wrong Qt version
G4 data files not found at runtime	Forget to source <code>geant4.sh</code>
Visualization window doesn't open	Install OpenGL-X11 and Qt-OpenGL dev packages
Segfault in <code>G4UImanager</code>	Merge file path wrong — use absolute path